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Unit 2 Lab Student Report
Collection and Examination
of Digital Evidence

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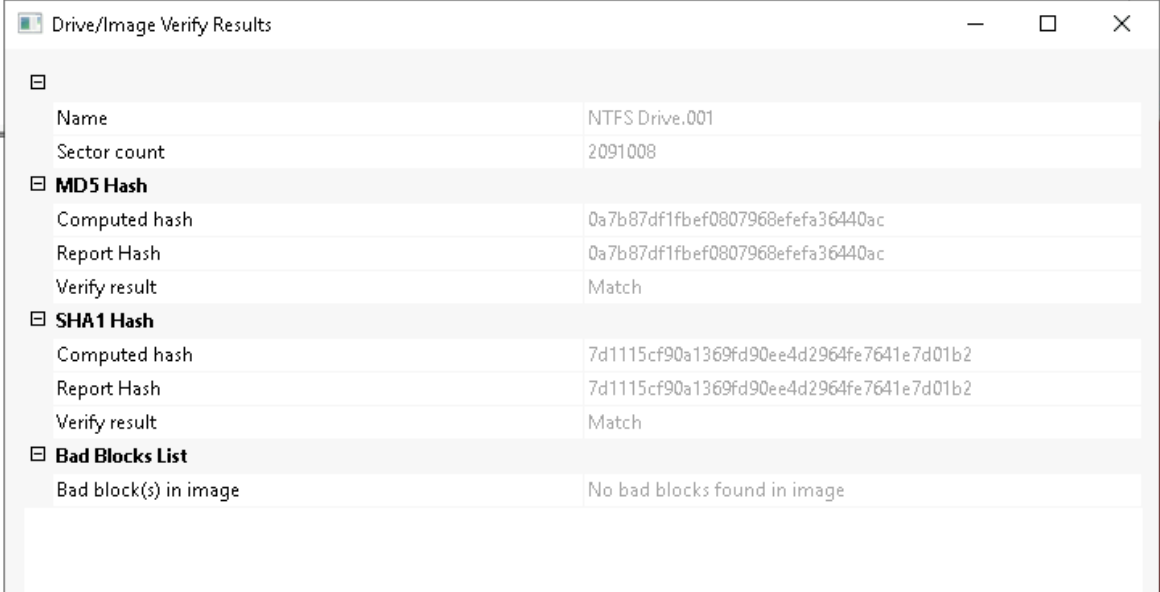
Introduction

The primary focus of this lab was to learn how to image and verify the images of NTFS drives, FAT32 drives, folders, and hash lists. FTK Imager was used to create images and HashCalc was used to verify the SHA1 and MD5 hashes. This was performed by first creating the appropriately named folders. We then opened FTK Imager, created logical drive disk images, selected the appropriate file, and ran and verified the hashes. HashCalc was then used to find the MD5 and SHA1 of the images to verify they matched FTK Imager. By using two sources, it confirmed that everything matched and a sound baseline was created.

It is important to note that volumes can be physical and logical to create images from. The physical volume is the drive in its entirety. In the case of this lab, we created logical volume images as we were only imaging a portion of the drive. Specifically, we imaged the named folders. We were not concerned with the entire drive for this lab, otherwise we would have created a physical volume image.

Screenshot 1 – Image of NTFS Drive

Figure 1: NTFS Drive/Image verification results

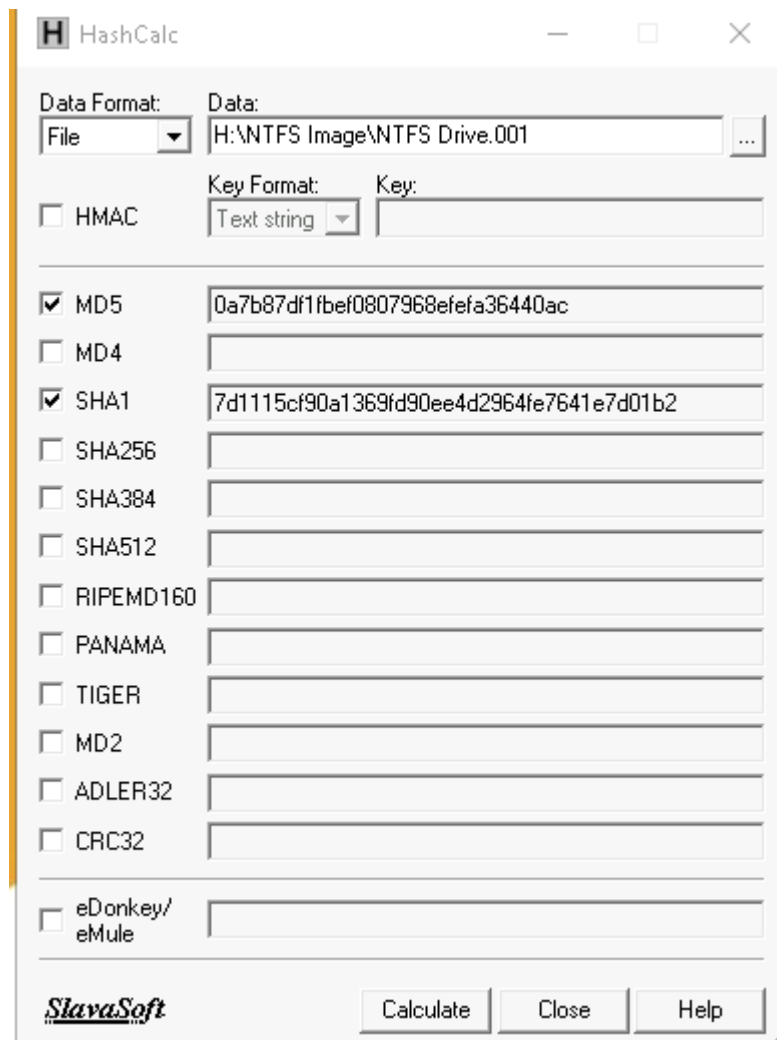


The screenshot shows a window titled "Drive/Image Verify Results" with a tree view on the left and a details pane on the right. The tree view has expandable sections for "Name", "Sector count", "MD5 Hash", "SHA1 Hash", and "Bad Blocks List". The details pane shows the corresponding values for the selected item, "NTFS Drive.001".

Drive/Image Verify Results	
Name	NTFS Drive.001
Sector count	2091008
MD5 Hash	
Computed hash	0a7b87df1fbef0807968efefa36440ac
Report Hash	0a7b87df1fbef0807968efefa36440ac
Verify result	Match
SHA1 Hash	
Computed hash	7d1115cf90a1369fd90ee4d2964fe7641e7d01b2
Report Hash	7d1115cf90a1369fd90ee4d2964fe7641e7d01b2
Verify result	Match
Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

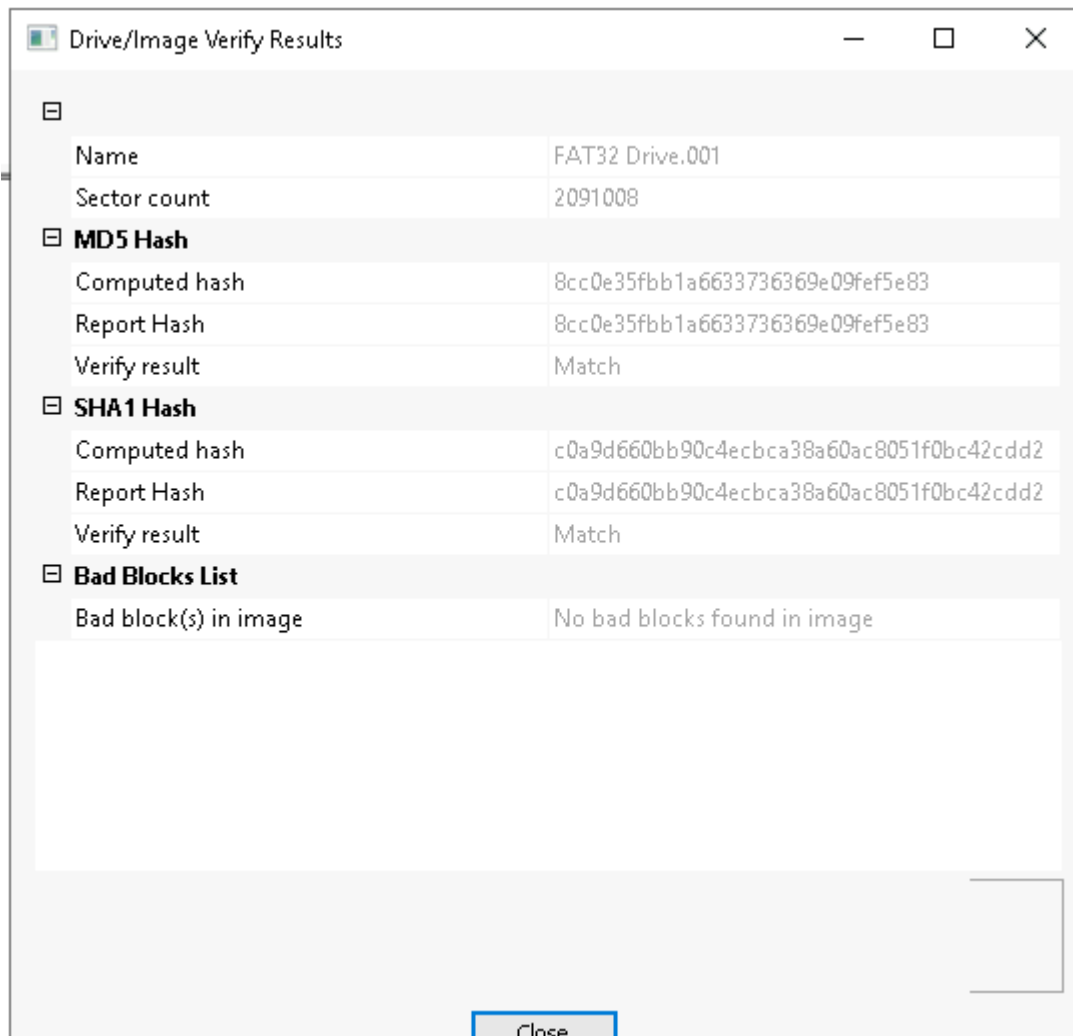
Screenshot 2 – Verify NTFS Drive Image

Figure 2: Results of NTFS Drive/Image in Hashcalc



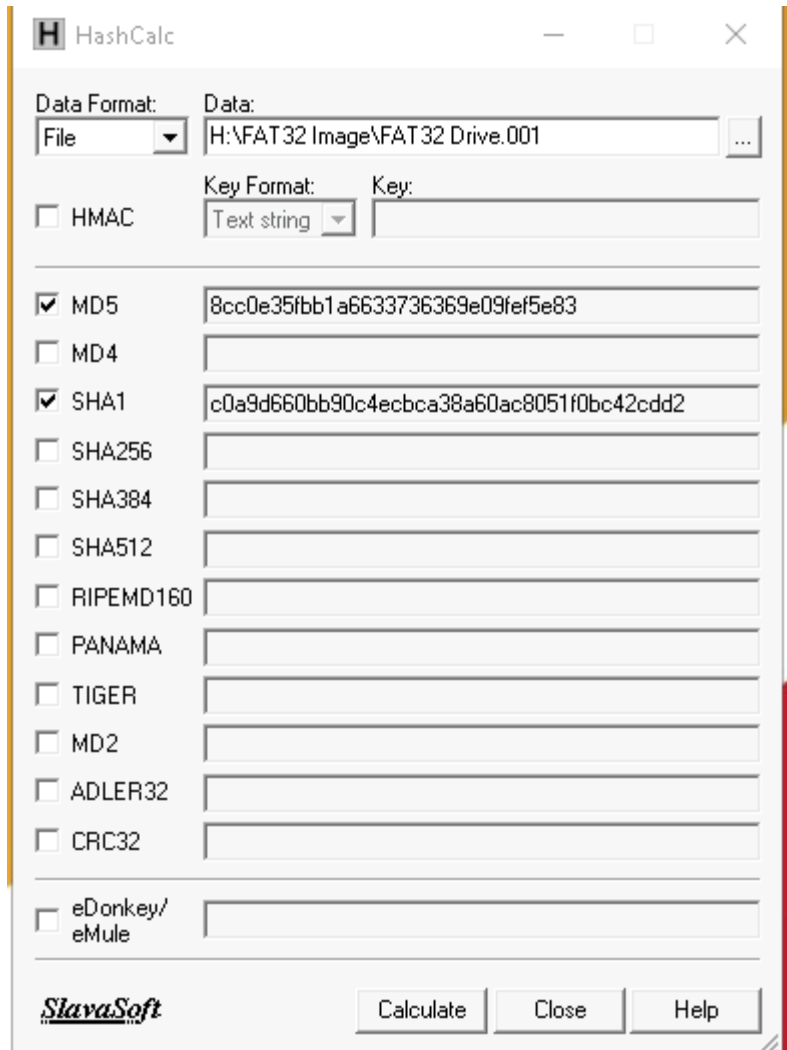
Screenshot 3 – Image FAT32 Image

Figure 3: FAT32 Drive/Image verification results



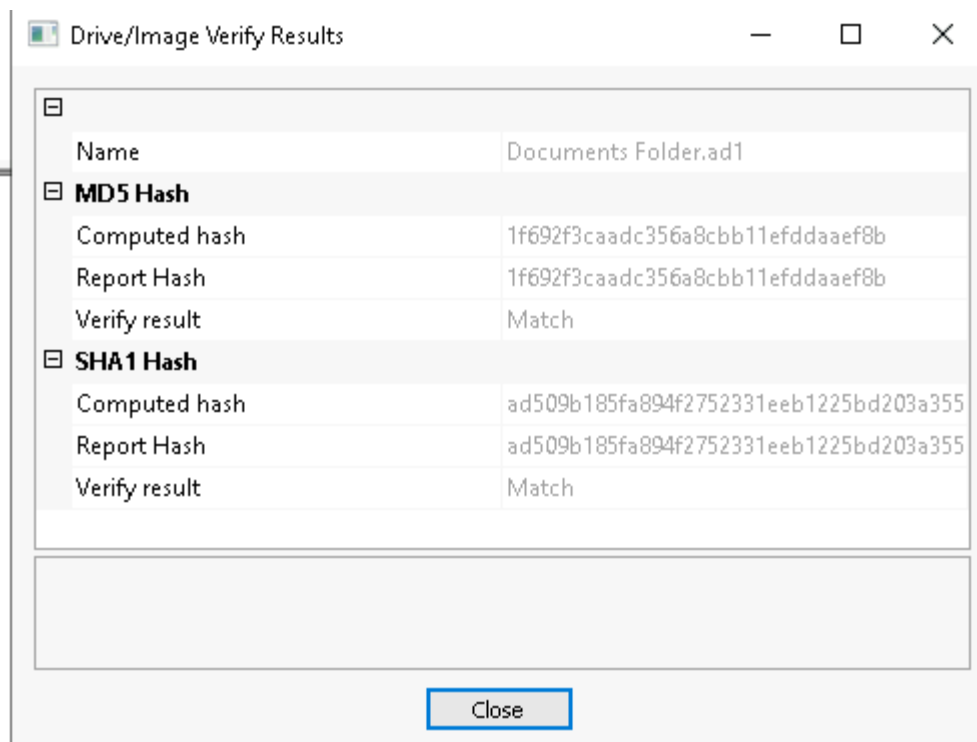
Screenshot 4 – Verify FAT32 Drive Image

Figure 4: Results of HashCalc hash verification on FAT32 Drive/Image



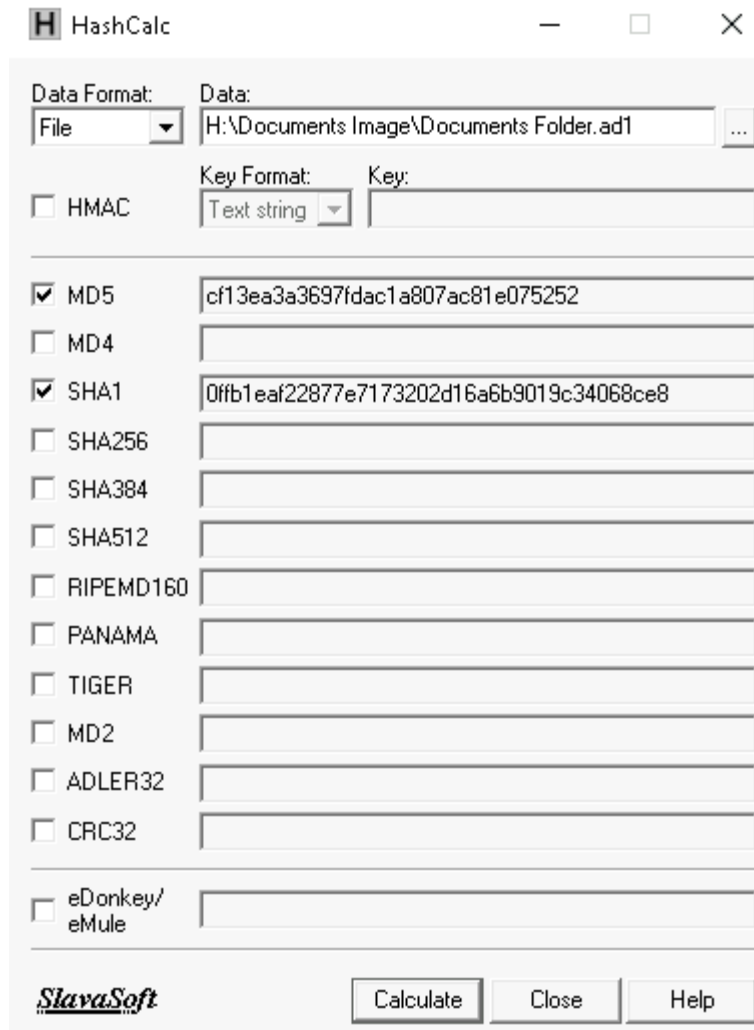
Screenshot 5 – Image a Folder

Figure 5: Results of Documents Folder FTK Image hash verification



Screenshot 6 – Verify Folder Image

Figure 6: Results of Document folder image in HashCalc



Screenshot 7 – Create Hash List of Folder Contents

Figure 7: Results of Document Image Hash List created in FTK Imager

A1			MD5
	A	B	C
1	MD5	SHA1	FileNames
2	52a32a6a4465a41128e538534a66ad14	b514dc854be9857664760d154a3779549a76ca92	Documents Folder.ad1\Default.rdp
3	ecf88f261853fe08d58e2e903220da14	f72807a9e081906654ae196605e681d5938a2e6c	Documents Folder.ad1\desktop.ini
4	c487c9f7d97e264ae28b137a3f5398b0	ef4c405eff2a985b8f7abe0049f142c333530a5	Documents Folder.ad1\EnCase\EAJobs\Jobs.ini
5	f3b25701fe362ec84616a93a45ce9998	d62636d8caec13f04e28442a0a6fa1afeb024bbb	Documents Folder.ad1\EnCase\Logs\Console1.txt
6	f3b25701fe362ec84616a93a45ce9998	d62636d8caec13f04e28442a0a6fa1afeb024bbb	Documents Folder.ad1\EnCase\Logs\Console2.txt
7	f3b25701fe362ec84616a93a45ce9998	d62636d8caec13f04e28442a0a6fa1afeb024bbb	Documents Folder.ad1\EnCase\Logs\Console3.txt
8	f3b25701fe362ec84616a93a45ce9998	d62636d8caec13f04e28442a0a6fa1afeb024bbb	Documents Folder.ad1\EnCase\Logs\Console4.txt
9	44b595fdb5e9574661bd8d9dae701b38	40c0c4bf49c0977c1a7100ac58afb485c079f4e4	Documents Folder.ad1\gruyere-code\data.py
10	5c0bad70adbb1589b3356a1bfc78a8a1	84e8ea842f291d9d68720578de3c7bb7171c2	Documents Folder.ad1\gruyere-code\gruyere.py
11	310d58de68fa40cf9e2875e39447b9d9	696e9d858ed7081ec2ddab9d3af41bb0b5328468	Documents Folder.ad1\gruyere-code\gtl.py
12	0ada2874322613d03b337b4c70289ef4	63b3c25166941b13c9ae7d949caafa8d3869e495	Documents Folder.ad1\gruyere-code\README
13	9e35a8b0c2b4113e50e066a378a1d2ff	5c0c78938e7a8aff442bceb6ac4aa4108dd81936	Documents Folder.ad1\gruyere-code\resources\base.css
14	422fa4295609b1f4acd99ffe9e98b706	a9d3d14b2625c9de568026936e50653bead6f1c	Documents Folder.ad1\gruyere-code\resources\cheese.png
15	c2a0da845a6e7417a9093ed84edb4f0	0e3dd1523125f13971d91e8b2279420f32450af4	Documents Folder.ad1\gruyere-code\resources\dump.gtl
16	bf7cc181b4eac44a9aa3761d6eb7ea8e	c863757372f5f155c77ea9b8af472253a60213ae	Documents Folder.ad1\gruyere-code\resources\editprofile.gtl
17	ee2029c60272be66ee95e91d2ec94702	39c6653b8fb02b8a4195db36e5253a15877d960c	Documents Folder.ad1\gruyere-code\resources\error.gtl
18	83d7c869258b8be83bdb66a997552b6f	cd1d1793dad579f217b331d412aa1fccbfe454a	Documents Folder.ad1\gruyere-code\resources\favicon.ico
19	d2b8e967d9f46f9be0a61a7efd9cf787	13606b41308cd55729037bebf74f616f8e27bf24	Documents Folder.ad1\gruyere-code\resources\feed.gtl
20	e1229ce9a9ae8f4f4e422c9e31cd48d1	785afb508af6ae3f0fa532cfa193e28094ee9ebf	Documents Folder.ad1\gruyere-code\resources\home.gtl
21	ce2231d3ab1cf3b490c9200c3857bc83	68858c6bdf15a301a376b25544883bab11cb1e79	Documents Folder.ad1\gruyere-code\resources\lib.js
22	fba414312677a729b0d31fe50aebdba6	27b03baf54c6e684a464a3bf57fba8b46a6b1e6d	Documents Folder.ad1\gruyere-code\resources\login.gtl
23	ffb0c1612014a2317f907998ad4cf50e	19ca1f46771448f7b586128b6ea5587957c7bf21	Documents Folder.ad1\gruyere-code\resources\manage.gtl
24	c4c822165241b29d136c1281fbc468f1	03edc987b12d774f11f49c1128fceac409ee63a	Documents Folder.ad1\gruyere-code\resources\menubar.gtl
25	5c52d03c19b64ccb02d3214d148804f7	fa500e4391f4314199c68316627de040f3e85e6f	Documents Folder.ad1\gruyere-code\resources\newaccount.gtl
26	4d9e7372393b34ed0d0c989943ebfbc6	a29b41c5a4c4cd98ffe2421b7a19f7aacd0c02a6	Documents Folder.ad1\gruyere-code\resources\newsnippet.gtl
27	6d7d656d84c7618c5cd4056150582802	2fd74051e16621d932bdb1098076c1fd37b4d650	Documents Folder.ad1\gruyere-code\resources\showprofile.gtl
28	023117606d638b596d16c102a0140eb3	b94d675aa994b80415f45b884721012534a230d3	Documents Folder.ad1\gruyere-code\resources\snippets.gtl
29	5b01fd2d9c400da49bd4c9d759a98b8b	9b078668785e2f697ebcf37fb585dd1656b94cbc	Documents Folder.ad1\gruyere-code\resources\upload.gtl
30	c5f35561411d444a4a2cde9f6d50921cd	ed2c16d73267887e2b0bfa1c6cbcf314844be212	Documents Folder.ad1\gruyere-code\resources\upload2.gtl
31	9d59ab11ae091c101c0092e44056c47e	489c546d5a20023e9b0a02d7fba524ef90824c7a	Documents Folder.ad1\gruyere-code\sanitize.py
32	4e6415e549a08a3e0ba13cd99a782436	97f05cfd8b65abbea5bfad3c685f205a959e63be	Documents Folder.ad1\gruyere-code\secret.txt
33	8bcccf174a518c601a27db3173aa5f89	41a9277cb37e0fb68e939552ba5e25dbc4b91400	Documents Folder.ad1\gruyere-code\stored-data.txt

Screenshot 8 – Hash Documents Image Hash List

Figure 8: Results of Document Image Hash List.csv in HashCalc

HashCalc

Data Format: **File** Data: H:\Documents Image\Documents Image Hash List.cs ...

☐ HMAC Key Format: Text string Key:

☒ MD5 1c64a3832c120908d60a036ec4a14df4

☐ MD4

☒ SHA1 8f551819974aacad0d3bb491e3eb184eed36cc0d

☐ SHA256

☐ SHA384

☐ SHA512

☐ RIPEMD160

☐ PANAMA

☐ TIGER

☐ MD2

☐ ADLER32

☐ CRC32

☐ eDonkey/
eMule

SlavaSoft Calculate Close Help

Conclusion

In the world of forensics, the golden rule is hashing. Before the investigation can be done and immediately after, the analyst must hash the image to ensure nothing was tampered and the image integrity was upheld. For example, if the computer hard drive contained critical evidence for a criminal case, the defense would check to see if the forensic analyst followed every rule or not. If the defense found that hashes were not taken, that evidence could be dismissed. This is why this lab is so crucial to the education of forensic analysts.

The reading was extremely beneficial for understanding EnCase file formats compared to FTK Imager. Additionally, speaking to a full-time forensic analyst assisted in understanding that E01 is far more standard of a file format than AD1. This is something that needs to be understood as a forensic analyst. The tools and file types an analyst uses must work together. E01 files have the benefit of a cyclical redundancy check that adds a file integrity verification as the image is being processed (Johansen, 2020). Additionally, it includes the MD5 hash at the end so that there are multiple integrity checks through the process.

The video by Hansani Vihanga covered the difference between live and static acquisitions well. It is critical that the individual conducting the live acquisition on site using write-block hardware and makes sure no one turns off the machine (Vihanga, 2021). A lot of critical data can be stored in cache or RAM that can be lost if the machine is turned off. Analysts as well as anyone on scene should be familiar with the order of volatility. The first responders on scene should ensure that no systems are turned off or put in sleep in case they have password protection.

I did not face any challenges during this lab. The guide was extremely detailed and provided clear instructions. It provided a great introduction to the process of creating images. Overall, this was an excellent introductory lab.

APA References

Johansen, G. (2020). *Digital forensics and incident response* (2nd ed.). Packt Publishing.

Vihanga, H. (2021, December 27). *Digital forensics: Acquisition and processing* [Video]. YouTube.

<https://www.youtube.com/watch?v=F5tv4c-f2Bw>